

PROMPT ENGINEERING EXAM QUESTION PAPER

RTCFR | Prompt Chaining | CoT | ToT | ReAct | Multi-Agent Prompting | Prompt Injection | Reverse Prompting

Student Name		Batch / Class	
Date		Time	
Total Questions	50	Total Marks	90 Minutes
Subject	Prompt Engineering	Exam Type	100
Instructions	Write clear, practical answers. No answer key is included in this document.		Written Practice

General Instructions

- Attempt all questions.
- Use RTCFR wherever required: Role, Task, Context, Few-shot, Response format.
- For prompt-writing questions, write practical prompts that can be directly used in ChatGPT, Claude, Gemini, or AI agents.
- Avoid writing answer keys or explanations outside the answer space.

Section A: Multiple Choice Questions

20 Questions × 1 Mark = 20 Marks

1. What does RTCFR stand for?

- ☒ A. Role, Task, Context, Few-shot, Response
- ☐ B. Reasoning, Tool, Context, Format, Result
- ☐ C. Role, Testing, Chain, Format, Response
- ☐ D. Read, Think, Create, Final, Review

Answer: A

2. In RTCFR, "Role" means:

- ☐ A. The final answer format
- ☒ B. The expert identity or perspective given to AI
- ☐ C. The number of examples in the prompt
- ☐ D. The user account type

Answer: B

3. Which one is a strong Role prompt?

- ☐ A. Tell me about marketing
- ☒ B. Act as a senior digital marketing strategist
- ☐ C. Give me answer fast
- ☐ D. Make it better

Answer: B

4. In RTCFR, "Task" means:

- ☐ A. Background information
- ☐ B. The exact work AI should perform
- ☒ C. The output design only
- ☐ D. A safety warning

Answer: B

5. Which is the clearest Task statement?

- ☐ A. AI
- ☐ B. Explain AI agents
- ☒ C. Create a 5-minute YouTube script explaining AI agents for Tamil business owners
- ☐ D. Help me with content

Answer: C

6. Context is important because:

- ☐ A. It helps AI guess randomly
- ☐ B. It reduces relevance
- ☒ C. It gives background information and improves accuracy
- ☐ D. It always makes the answer shorter

Answer: C

7. Few-shot prompting means:

- ☒ A. Giving one or more examples to guide the output
- ☐ B. Asking AI to answer in one line
- ☐ C. Asking AI to ignore examples
- ☐ D. Giving only a role

Answer: A

8. Response format in RTCFR means:

- ☒ A. How the final output should be structured
- ☐ B. The name of the AI model
- ☐ C. The system speed

D. The number of users

Answer: 4

9. Which is an example of a response format instruction?

- A. Act as a teacher
- ☒ B. Give the output in table format
- C. This is for students
- D. Use the internet

Answer: B

10. Prompt chaining means:

- A. Writing one long prompt only
- ☒ B. Splitting a big task into multiple connected prompts
- C. Removing context from the prompt
- D. Asking unrelated questions

Answer: B

11. Sequential prompt chaining means:

- A. Each step depends on the previous output
- B. All agents debate each other
- C. AI ignores previous answers
- D. Prompt is written backward

Answer: _____

12. CoT stands for:

- ☒ A. Chain of Thought
- B. Code of Testing
- C. Context of Tool
- D. Create Output Template

Answer: A

13. CoT prompting is mainly used for:

- ☒ A. Step-by-step problem solving and reasoning
- B. Image compression
- C. File storage
- D. Account login

Answer: A

14. ToT stands for:

- A. Task of Thinking
- ☒ B. Tree of Thoughts
- C. Tool of Testing
- D. Tone of Text

Answer: B

15. Tree of Thoughts prompting is useful when:

- ☒ A. There are multiple possible solution paths
- B. Only one-word answer is needed
- C. No reasoning is needed
- D. The answer must be random

Answer: A

16. ReAct prompting combines:

- ☒ A. Reasoning and Acting
- B. Reading and Copying
- C. Role and Task only
- D. Response and Title

Answer: A

17. ReAct prompting is useful for:

- ☒ A. Tool-based workflows and agentic tasks
- ☐ B. Only grammar correction
- ☐ C. Only image design
- ☐ D. Only translation

Answer: A

18. Multi-agent prompting means:

- ☐ A. One AI gives one answer only
- ☒ B. Multiple AI roles collaborate, critique, or debate
- ☐ C. AI removes all roles
- ☐ D. User gives no context

Answer: B

19. Prompt injection means:

- ☒ A. A malicious or unwanted instruction tries to override the original instructions
- ☐ B. A prompt with examples
- ☐ C. A prompt with table format
- ☐ D. A prompt with keywords

Answer: A

20. Reverse prompting means:

- ☒ A. Asking AI to create a prompt from a given output or goal
- ☐ B. Asking AI to delete the prompt
- ☐ C. Asking AI to avoid examples
- ☐ D. Asking AI to answer without a task

Answer: A

Section B: 2-Mark Short Answer Questions

15 Questions × 2 Marks = 30 Marks

21. Define RTCFR in simple words. (2 Marks)

Role, Task, Context, Format, & Rules.
5 part checklist used to write clear and helpful instructions for AI chatbots.

22. Write the five components of RTCFR. (2 Marks)

Role, task, context, format, rules.

23. Why is Role important in a prompt? (2 Marks)

Deciding who you want the AI to be.

24. Write two examples of good Role prompts. (2 Marks)

If you just ask "What is AI or computer" ×
"What is MAC Studio" or specify the requirement.

25. What is the difference between Task and Context? (2 Marks)

It's very simple Task stand for to-do
No answer key included | Students can write answers in the spaces provided

Contest stand for information background information

26. Write one example of a clear Task prompt. (2 Marks)

write me 3-paragraph introduction for my blog.

27. What is Few-shot prompting? (2 Marks)

Examples ->

28. Write one example of Few-shot prompting for YouTube titles. (2 Marks)

Zero-shot "write a polite email declining a job offer."

29. What is Response format in prompting? (2 Marks)

Report; Reporting

30. Write three examples of response formats. (2 Marks)

Response email to thank you for the job offering
write in conversational style
Email format to be - - - -

31. What is prompt chaining? (2 Marks)

Prompts guide AI through a series of logical steps to complete a task.

32. What is the difference between Sequential prompting and normal prompting? (2 Marks)

Normal prompting completes a task in a single input-output turn, while sequence prompting breaks a complex task into a step-by-step

33. What is the purpose of CoT prompting? (2 Marks)

To improve an AI's reasoning and accuracy on complex task by encouraging it to break down problem and generate intermediate.

34. What is prompt injection? Give one simple example. (2 Marks)

Prompt injection is a security vulnerability where an attacker uses crafted input.

35. What is reverse prompting? Where can it be used? (2 Marks)

Ask AI to generate the RTCFR form for us.

Section C: Write Prompts

10 Questions × 3 Marks = 30 Marks

36. Write an RTCFR prompt to create a YouTube video script about AI Agents for Tamil beginners. (3 Marks)

Expected elements: Role, Task, Context, Few-shot example, Response format

1. Create a Youtube video script about AI agent for 5 min
2. Think you are Director and a tech person to plan
3. script for AI agent how it works and how to train
4. step-by-step in a table format.
5. script should be in for ~~best~~ Tamil beginners.

37. Write a prompt to create a professional email template for a support ticket system. (3 Marks)

Scenario: A student submits a doubt through Google Forms. Data goes to Google Sheets. Email should go to the support team.

write email content for a support ticket for the student request.
All the Google forms Data should go to google sheets and also cc the supporting team.

38. Write a Sequential Prompt Chain for creating a YouTube video. (3 Marks)

Include: generate topic ideas, select best topic, create title, create script, create description and tags.

Topic → Research - outline → script → visuals
→ Voiceover → Thumbnail → Final Video package

39. Write a CoT-style prompt for solving a business problem. (3 Marks)

Scenario: A small business owner wants to automate customer support using AI.

A CoT-style prompts guides the AI to break the business problem into smaller steps, analyze each factor, identify the root cause, and then provide a structured recommendation.

40. Write a Tree of Thoughts prompt for choosing the best AI tool for a company. (3 Marks)

Compare: ChatGPT, Claude, Gemini, Local LLM.

Tree of Thought means exploring multiple possible or decision paths, comparing them against.

the requirement, eliminating weaker options and selecting the best solution.

41. Write a ReAct prompt for an AI agent that uses tools. (3 Marks)

Scenario: The agent should check Google Sheets, summarize new leads, and draft a Gmail follow-up.

ReAct prompting combines reasoning and action. The AI first determines what it needs to do, uses the appropriate tool, observes the result, and then decides the next action until it has provided the final answer.

42. Write a Multi-agent AI prompt for planning a product launch. (3 Marks)

Use at least three agents: Marketing expert, Sales expert, Product manager.

Multi-agent AI prompt divides a complex task among multiple specialized AI agents. Each agent handles a specific area, shares its findings with the other agents, and a coordinator agent combines everything into one final solution.

43. Write a prompt injection defense instruction for a system prompt. (3 Marks)

Scenario: The AI should not follow user instructions that ask it to ignore previous instructions or reveal hidden instructions.

Prompt injection defense ensure that instructions from users, documents, websites, or tool outputs cannot override the system's trust instructions. The AI treats external content as untrusted data, validates tool actions, protects sensitive information, and follows the instruction hierarchy.

44. Write a reverse prompting request. (3 Marks)

Scenario: You saw a very good LinkedIn post and want AI to create the prompt that could generate similar posts.

Reverse prompting is the process of starting with a desired AI output and working backward to create a prompt that can produce a similar result.

45. Write a prompt using at least 5 powerful prompt keywords. (3 Marks)

Use keywords like: Analyze deeply, Step-by-step, Practical examples, Checklist, Optimize for clarity.

1. Act as expert
2. Think step-by-step
3. Break it down
4. Give practical examples
5. Step-by-step guide

Section D: Identify RTCFR Elements

5 Questions × 2 Marks = 10 Marks

46. Identify Role and Task from this prompt: "Act as a senior YouTube strategist. Create 10 viral YouTube titles for a Tamil video about n8n automation." (2 Marks)

Role:- Act as a senior youtube strategist

Task:- create 10 viral youtube titles for a tamil video about n8n automation.

47. Identify Context from this prompt: "Act as a career coach. Create a 30-day learning plan for a beginner who knows basic Python and wants to become an AI automation developer." (2 Marks)

Who knows basic python and wants to become an automation developer

48. Identify Few-shot from this prompt: "Generate Tamil YouTube titles. Style examples: 1. AI Tools வைத்து Business Automation பண்ணலாம் 2. ChatGPT கத்துக்கணுமா? Start Here 3. n8n Automation Full Beginner Guide." (2 Marks)

all the examples which make here is a Few-shots

49. Identify Response format from this prompt: "Give the output in a table with columns: Day, Topic, Task, Tool, Expected Output." (2 Marks)

Give the output in a table with columns.

50. Identify all RTCFR elements from this prompt: "Act as a senior AI trainer. Create a beginner-friendly lesson plan on Prompt Engineering. My audience is Tamil business owners who are new to AI. Example style: simple English with Tanglish examples. Give output in table format with lesson title, explanation, demo, and practice task." (2 Marks)

Response

Viewsshots

c

Section E: Prompt Keywords Practice

Optional Revision Section

No answer key included | Students can write answers in the spaces provided

Write any 10 powerful prompt keywords you can use to get better AI output:

1. Act as an ~~expert~~ expert
2. Think step by step
3. Break it down
4. Give practical examples
5. step-by-step guide
6. checklist format
7. improve this
8. optimize for clarity
9. give me 5 variations
10. Output in JSON

--- End of Question Paper ---

5